

Yusheng (Jason) Zhang

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PROFESSIONAL PROFILE

Mechanical Engineering PhD candidate with experience in finite element analysis, mechanical testing, and computational modeling to evaluate performance and failure behavior of device-scale systems. Background includes simulation and experimental validation of stress, deformation, and fluid driven interactions in biomedical devices. Skilled in analyzing complex datasets and translating technical findings into clear, defensible conclusions for both technical and nontechnical audiences.

EDUCATION

Global PhD Fellow, New York University (New York, NY & Abu Dhabi, UAE)

Doctor of Philosophy in Mechanical Engineering (PI: Prof. Khalil Ramadi, [Ramadi Lab](#)) Expected May 2027, GPA: 4.0/4

Northeastern University (Boston, MA)

Master of Science in Mechanical Engineering (Concentration in Thermofluids)

May 2023, GPA: 3.917/4

Bachelor of Science in Mechanical Engineering

Aug 2022, GPA: 3.682/4

RESEARCH EXPERIENCE

Tunable Ingestible Mechanical Systems for GI Applications, New York University

Sep 2023 – Present

- Designed and evaluated ingestible mechanical systems for gastrointestinal deployment, integrating CAD modeling, fluid structure considerations, and experimental validation to assess device performance under physiological constraints
- Performed computational simulations, benchtop, and in vivo testing to analyze system behavior, optimize retention mechanisms, and identify performance limitations and potential failure modes
- Developed rapid prototyping workflows using additive manufacturing to iterate on device design, geometry, and mechanical response
- Synthesized experimental and computational results to evaluate design tradeoffs and guide engineering decisions
- Communicated technical findings through presentations and reports to multidisciplinary research teams

Intracerebral Microfluidic Catheter Design and Evaluation, New York University

Dec 2024 – Sep 2025

- Conducted finite element analysis (FEA) to evaluate stress distributions and structural integrity of novel catheter geometries, comparing performance against conventional designs
- Identified mechanical advantages through analysis of von Mises stress under prescribed displacement conditions, demonstrating improved resilience and reduced stress concentrations
- Designed and executed mechanical testing protocols (Instron) to validate simulation predictions and characterize failure behavior
- Correlated experimental results with computational models to support design validation and performance assessment

Flexible Intestinal Bioelectronic Device Development, New York University

Jul 2024 – Present

- Performed mechanical and compliance analysis of flexible bioelectronic systems to ensure consistent tissue contact and functional performance
- Conducted ex-vivo testing and force measurements to evaluate device-tissue interaction and mechanical stability
- Applied simulation and experimental methods to optimize device design under dynamic physiological conditions

Combustion and Fluid System Analysis of Air-Fuel Mixtures, Northeastern University

Apr 2022 – May 2023

- Conducted experimental studies on hydrogen-methane-air mixtures to evaluate combustion behavior under varying equivalence ratios
- Developed computational models in MATLAB and Python to calculate laminar burning velocities and analyze fluid and thermal system behavior
- Integrated experimental and computational results to validate models and identify trends in fluid driven system performance

WORK EXPERIENCE

Systems Engineer

Fresenius Medical Care/NxStage (Lawrence, MA)

Jul – Dec 2021

- Analyzed fluid system performance in a dialysis device prototype by evaluating pressure data and flow control parameters, identifying sources of inefficiency and variability
- Designed and executed testing protocols to diagnose system behavior and improve reliability of fluid management processes
- Reduced calibration error by 75% through systematic evaluation of sensor data and redesign of calibration procedures
- Synthesized experimental data to support engineering decisions and improve overall system performance

Supplier Engineer

Insulet Corporation (Acton, MA)

Jul – Dec 2020

- Designed and prototyped mechanical fixtures using CAD and additive manufacturing to support device testing and validation
- Conducted mechanical testing (Instron) and analyzed force-response data to evaluate component performance
- Reviewed supplier technical drawings and validation data to ensure compliance with engineering specifications
- Developed documentation and validation reports summarizing testing results and engineering assessments

Sales Representative

Dell Retailer (Shenzhen, China)

Jul – Aug 2018

- Assessed customer needs and advised customers based on understanding of their needs
- Negotiated pricing with customers and processed sales
- Collected customer feedback through phone surveys and created reports to share information on services and products

SKILLS

Computer Applications: SolidWorks, COMSOL, AutoCAD, MATLAB, Python, Prism, Adobe Illustrator, Agentic AI

Additive Manufacturing (3D Printing): Stratasys J750/J850, Boston Micro Fabrication (BMF) S240

Cleanroom: 3D Nano Printing Systems – Nanoscribe Photonic Professional GT & Nanoscribe Quantum x Bio

Imaging: Skyscan 1276 Micro-CT System, ThermoFischer Quanta 3D Scanning Electron Microscope (SEM)

Measurement Devices: Instron, OGP SmartScope

Languages: Fluent in Mandarin and Cantonese, Intermediate in Spanish

ACHIEVEMENTS

Winner, [Merck Innovation Cup 2025](#)

- Initially selected from a highly competitive global applicant pool of PhD candidates, postdoctoral researchers, and industry professionals to join 40 participants in the program in Darmstadt, Germany
- Developed AI-driven manufacturing concept in a global multidisciplinary team

Winner, New York University (Abu Dhabi) 3-Minute Thesis Competition ([GradSlam](#))

- Clinched the Grand Prize at GradSlam, a prestigious campus-wide 3-minute research presentation competition for graduate students and postdoctoral researchers

Best Presentation, 1st UAE Biomaterials, Biofabrication, and Bioengineering Conference ([U3B](#))

- Selected as 1 of 11 early career researchers to present at the inaugural U3B conference, the first of its kind in the United Arab Emirates and the Middle East and North Africa (MENA) region
- Presentation on a bio-inspired ingestible platform for prolonged gastrointestinal biosensing recognized as a best presentation in the Rapid Fire Presentation section

LEADERSHIP

Secretary/Tandon Student Representative | NYU Graduate Student Council

Sep 2024 – Present

- Coordinated student initiatives and organized events to foster collaboration and engagement across diverse groups